

$$1) a) (A \cup B)^c = (A^c \cap B^c) \rightarrow \text{Seja } x \in (A \cup B)^c$$

$$b) (A \cap B)^c = A^c \cup B^c \rightarrow \text{Seja } x \in (A \cap B)^c$$

$$a) \text{ Seja } x \in A^c \cap B^c$$

$$\hookrightarrow x \notin A \text{ e } x \notin B$$

$$\hookrightarrow x \notin A \cup B$$

$$\hookrightarrow x \in (A \cup B)^c$$

$$A^c \cap B^c \subseteq (A \cup B)^c$$

$$(A \cup B)^c = A^c \cap B^c$$

$$b) \text{ Seja } x \in (A \cap B)^c$$

$$\hookrightarrow x \notin A \cap B$$

$$\hookrightarrow x \notin A \text{ ou } x \notin B$$

$$\hookrightarrow x \in A^c \text{ ou } x \in B^c$$

$$\hookrightarrow x \in A^c \cup B^c$$

$$(A \cap B)^c = A^c \cup B^c$$

$$2) a) (\bigcup_{i \in I} A_i)^c = \bigcap_{i \in I} A_i^c$$

$$x \in (\bigcup_{i \in I} A_i)^c$$

$$\hookrightarrow x \notin \bigcup_{i \in I} A_i$$

$$\hookrightarrow x \notin A_i \text{ para todo } i \in I$$

$$A_i \in A_i^c \quad \text{ " " "}$$

$$\hookrightarrow x \in \bigcap_{i \in I} A_i^c$$

$$b) (\bigcap_{i \in I} A_i)^c = \bigcup_{i \in I} A_i^c$$

$$x \in \bigcap_{i \in I} A_i$$

$$\hookrightarrow x \in A_i \text{ para todo } i \in I$$

$$x \in A_i^c \subseteq \bigcup_{i \in I} A_i^c$$

$$3) (A \cup B) \cap C = A \cap (B \cap C)$$

Associativa da uniao

Nao muda o resultado

$$(A \cap B) \cup C = A \cap (B \cup C)$$

Associativa da intersecao

$$4) (A \cup B) \cap C = (A \cap C) \cup (B \cap C)$$

$$x \in (A \cup B) \cap C$$

$$\hookrightarrow x \in A \cup B \text{ e } x \in C$$

$$x \in A \cap C \text{ ou } x \in B \cap C$$

$$x \in (A \cap C) \cup (B \cap C)$$

$$(A \cap B) \cup C = (A \cup C) \cap (B \cup C)$$

$$x \in (A \cap B) \cup C$$

$$\hookrightarrow x \in A \cap B \text{ ou } x \in C$$

$$\hookrightarrow x \in A \cup C \text{ e } x \in B \cup C$$

$$\hookrightarrow x \in (A \cup C) \cap (B \cup C)$$

Master Que

$$A \setminus B \neq B \setminus A$$

$$A = \{2\}$$

$$B = \{1, 2, 3\}$$

$$B \setminus A = \{3\}$$

$\neq$

$$A \setminus B = \emptyset$$

$$A \setminus B = A \cap B^c$$

$$\hookrightarrow \emptyset$$

$$x \in A \cap B^c$$

$$x \in A \cap B^c$$

$$x \in B^c$$

$$x \in A \cap B^c$$

$$x \in B$$

$$\hookrightarrow x \in A \cap B^c$$

$$\hookrightarrow x \in A \cap B$$